MILITARY SPECIFICATION
CONTROL, GUNNERS, LASER RANGING: 10559221

1. SCOPE

1.1 This specification covers one type of electronic unit known as Control, Gunners, Laser Ranging: 10559221 (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10559221	Control, Gunners, Laser Ranging

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PACKAGING DATA SHEET

10559221 Packaging of Control, Gunners, Laser Ranging: 10559221

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The gunners control shall be manufactured in accordance with Drawing F10559221 and drawings pertaining thereto.

3.2 General specification. - The gunners control shall meet the following requirements, where applicable, of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Part identification and marking

3.2.1 Electronic workmanship. - The gunners control shall meet the workmanship requirements of Drawing F10559221 and drawings pertaining thereto.

3.3 Performance. - Unless otherwise specified, the gunners control shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit ($^{\circ}$F) and 90°F.

3.3.1 Load, power, and signals. - The gunners control shall perform as specified herein when the load, power, and signals, of Table I are applied as specified.

TABLE I

ITEM	CONDITION	CHARACTERISTICS	CONNECTIONS
1.	Load:		
1.1	Resistor	120 ohms ±5%	Connected between J3-V and J3-X

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TABLE I (continued)

ITEM	CONDITION	CHARACTERISTICS		CONNECTIONS
2.	Power sources:	Tolerance	Maximum p-p ripple	Unless otherwise specified, applied between following pins of J1:
2.1	+5 Vdc	±0.2 Vdc	25 mV	W(+) and X(-)
2.2	+15 Vdc	±0.7 Vdc	25 mV	V(+) and X(-)
3.	Signal sources:			
3.1	Digital type-A	Logical one: +4 ±1V Logical zero: +0.2 ±0.2V		Applied as specified in subsequent tables herein.
3.2	Digital type-C	Logical one: +19 ±1 Vdc Logical zero: Open circuit		

3.3.1.1 Switching continuity. - With no power or signals applied, an open or short-circuit condition shall be present between the terminals in Table II as specified for the various switch positions.

TABLE II

SWITCH	SWITCH POSITION		TERMINALS (PINS of J1)			
			p & X	q & X	r & X	s & X
RANGE RETURN SELECTOR	1	Normal	O	NA	NA	NA
		Depress	S	NA	NA	NA
	2	Normal	NA	O	NA	NA
		Depress	NA	S	NA	NA
	LAST	Normal	NA	NA	O	NA
		Depress	NA	NA	S	NA
FEED		Normal	NA	NA	NA	O
		Depress	NA	NA	NA	S

Legend: S = short circuit; O = open circuit
(<0.3 ohm) (>1 megohm)
NA = Not applicable

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3.3.1.2 FEED and LAST indicators. - Upon application of the power specified in Table I, but with no signals applied, the indicator lamps marked "FEED" and "LAST" shall light.

3.3.1.3 Dimmer switch (S1). - Depressing the dimmer switch (S1) shall cause all indicator lamps to change either from bright to dim or from dim to bright.

3.3.1.4 Indicator illumination.

3.3.1.4.1 RANGE RETURN SELECTOR indicators. - With the logical one, digital type-A signal of Table I applied to J2-q, the indicator lamps marked "1", "2", and "LAST" shall light when the associated switches S2, S3, and S4, respectively, are depressed.

3.3.1.4.2 RANGE and RESET indicators. - With the appropriate digital signals of Table I applied as specified in Table III, the corresponding indicator lamps shall light as specified in Table III.

TABLE III

ITEM	INDICATOR ILLUMINATION	INPUT TERMINALS	
		J2-r	J2-Y
1	RESET	E	1
2	RANGE (White)	E	1
3	RANGE (Red)	E	0

Legend: E = logical one, digital type-C signal
0 = logical zero, digital type-A signal
1 = logical one, digital type-A signal

3.3.1.5 Switch selection logic. - With the logical one, type-A signal of Table I applied to J1-t, and the logical one, type-C signal of Table I applied to J2-r, the voltages at the output terminals in Table IV shall be as specified for the listed switch positions.

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TABLE IV

SWITCH	POSITION	OUTPUT TERMINALS		
		J2-s	J2-m	J3-n
RESET	Normal	NA	4.2 ±1V	0.2 ±0.2
	Depress	NA	0.2 ±0.2V	NA
RANGE	Normal	0.2 ±0.2V	NA	NA
	Depress	4.2 ±1V	NA	NA

3.3.2 Environmental.

3.3.2.1 Storage temperature. - The gunners control shall meet the requirements of 3.3.1 at ambient temperature (60°F to 90°F) after exposure and thermal stabilization at -80°F and +160°F.

3.3.2.2 Operating temperature. - The gunners control shall meet the requirements of 3.3.1 while exposed to and thermally stabilized at -40°F and +125°F, subsequent to which it shall meet the above requirements at ambient temperatures (60°F to 90°F).

3.3.2.3 Vibration. - The gunners control shall meet the requirements of 3.3.1 after being exposed to the following vibratory conditions:

(a) Amplitude:	0.5-inch maximum
(b) Acceleration:	4g maximum
(c) Crossover frequency:	12.5 Hertz
(d) Sweep time:	5 to 500 to 5 Hertz in 15 minutes. Two sweeps in each of the three mutually perpendicular axes.

3.3.2.4 Shock. - The gunners control shall meet the requirements of 3.3.1 after being exposed to a total of 6 shock impulses, one in each direction of the three mutually perpendicular axes. Each shock impulse shall be a sawtooth or half-sine wave with a time duration of 6 milliseconds. The peak amplitude of each shock impulse shall be 100g.

3.3.2.5 Immersion. - With the gunners control pressurized to 5 ±1 psig with nitrogen or air and submerged in water to a depth not to

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exceed 2 feet at a temperature of 110 ±10°F, there shall be no evidence of gas bubbles emanating from inside the unit during a 15-minute observation period.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. - The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three gunners controls. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables V, VI, and VII herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables V, VI, and VII shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests.- The requirements and tests in Table V shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE V
(100 percent inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101. Switching continuity	3.3.1.1	4.6.2
102. FEED and LAST indicators	3.3.1.2	4.6.3
103. Dimmer switch (S1)	3.3.1.3	4.6.4

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TABLE V (Continued)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
104. Indicator illumination	3.3.1.4	4.6.5
105. Switch selection logic	3.3.1.5	4.6.6

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling

4.4.1 General.- One gunners control shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table VI.

TABLE VI

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Applicable drawings - visual
302. General specification	3.2	MIL-F-13926 - visual

4.4.2 Environmental.- Three gunners controls shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table VII.

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TABLE VII

CONTROL TEST		REQUIREMENT	TEST PROCEDURE
303.	Storage temperature	3.3.2.1	4.6.7.1
304.	Operating temperature	3.3.2.2	4.6.7.1
305.	Vibration	3.3.2.3	4.6.7.2
306.	Shock	3.3.2.4	4.6.7.3
307.	Immersion	3.3.2.5	4.6.7.4

4.4.3 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted item has been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment. - Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test

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equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment shall reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment. - In conjunction with 4.5.2, the standard test equipment listed in Table VIII shall be utilized in performing the applicable test of 4.6.

TABLE VIII

NOMENCLATURE	DESCRIPTION
1. D-C Power Supply	Plus 5 ±.2 V Ripple: 25 mV p-p maximum
2. D-C Power Supply	Plus 15 ±.7 V Ripple: 25 mV p-p maximum
3. Ohmmeter	Range: 0 to 10 megohms Accuracy: ±5%
4. D-C Voltmeter	Range: 0 to 10 V Accuracy: ±1%
5. Signal Source	Capable of providing a d-c level of amplitude as specified in Items 3.1 and 3.2, Table I

4.6 Test methods and procedures

4.6.1 Test conditions. - Unless otherwise specified, the test conditions shall be in accordance with the "Test Facilities" requirements of MIL-F-13926.

4.6.1.1 Initial conditions. - Connect the load and power (except in 4.6.2) to the gunners control as specified in Table I.

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4.6.2 Switching continuity. - With no power or signals applied, measure for an open or short-circuit condition between the terminals in Table II as specified for the various switch positions. The open or short-circuit conditions shall be as specified in 3.3.1.1.

4.6.3 FEED and LAST indicators. - Apply the power, but not the signals, as specified in Table I. The indicator lamps marked "FEED" and "LAST" shall light as specified in 3.3.1.2.

4.6.4 Dimmer switch (S1). - With the indicator lamps lighted as in 4.6.3 and 4.6.5, depress the dimmer switch. All indicator lamps shall change either from bright to dim or from dim to bright as specified in 3.3.1.3.

4.6.5 Indicator illumination.

4.6.5.1 RANGE RETURN SELECTOR indicators. - With the logical one, digital type-A signal of Table I applied to J2-q, depress switches S2, S3, and S4. The associated indicator lamps shall light as specified in 3.3.1.4.1.

4.6.5.2 RANGE and RESET indicators. - Apply the appropriate digital signals of Table I as specified in Table III. The corresponding indicator lamps shall light as specified in 3.3.1.4.2.

4.6.6 Switch selection logic. - Perform the following:

(a) Apply the logical one, type-A signal of Table I to J1-t.
(b) Apply the logical one, type-C signal of Table I to J2-r.
(c) Measure the voltages at the output terminals specified in Table IV for the specified RESET and RANGE switch positions.

The output voltages shall be as specified in 3.3.1.5.

4.6.7 Environmental tests. - The test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 except that the control may be mounted on an inside surface of the test chamber for the temperature test and the tolerance requirements for vibration frequency measuring instruments shall not apply.

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4.6.7.1 Temperature test. -

NOTE: The rate of temperature change shall be less than 7°F per minute during temperature cycling.

(a) Place the gunners control in the test chamber and reduce the chamber temperature to -80°F.
(b) Soak the gunners control at this temperature for at least 5 hours with the unit not operating.
(c) Raise the chamber temperature to -40°F.
(d) Following thermal stabilization at -40°F, place the gunners control in operation and perform the tests specified in 4.6.2 through 4.6.6.
(e) Raise the chamber temperature to +160°F.
(f) Soak the gunners control at +160°F for at least 5 hours with the assembly not operating.
(g) Reduce the chamber temperature to +125°F.
(h) Following the thermal stabilization at +125°F, place the gunners control in operation and perform the tests specified in 4.6.2 through 4.6.6.
(i) Reduce the chamber temperature to ambient.
(j) Following stabilization at ambient temperature, place the gunners control in operation and perform the tests specified in 4.6.2 through 4.6.6.

The gunners control shall meet the requirements of 3.3.2.1 and 3.3.2.2.

4.6.7.2 Vibration. - With the gunners control firmly mounted to the vibration table, perform the following:

(a) Test in accordance with Method 514-I of MIL-STD-810 for category (f) equipment, except that the acceleration shall not exceed ±4g and the maximum excursion (double amplitude) shall be 0.5 inch.
(b) Sweep the frequency from 5 to 500 Hz and back to 5 Hz in a period of 15 minutes. Perform two sweeps in each of the three mutually perpendicular axes.
(c) Following vibration cycling, perform a visual-tactile examination for damage and loose parts.
(d) Place the gunners control in operation and perform the tests specified in 4.6.2 through 4.6.6.

The gunners control shall meet the requirements of 3.3.2.3.

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4.6.7.3 Shock. - With the gunners control rigidly mounted to the shock machine table, perform the following:

(a) Test in accordance with Method 516, Procedure IV, of MIL-STD-810, except that the gunners control shall be subjected to one shock in each direction of the three mutually perpendicular axes and the shock impulse shall be a sawtooth or half-sine wave with a duration of 6 milliseconds and a peak amplitude of 100g.
(b) After exposure to shock, perform a visual-tactile examination for damage and loose parts.
(c) Place the gunners control in operation and perform the tests specified in 4.6.2 through 4.6.6.

The gunners control shall meet the requirements of 3.3.2.4.

4.6.7.4 Immersion. - Connect the leak test port of the gunners control to a cylinder of nitrogen or air and pressurize it to 5 ±1 psig. Submerge the gunners control in water at 110 ±10°F to a depth not to exceed 2 feet and observe the gunners control for evidence of leakage (gas bubbles) for a period of 15 minutes. (Note: The gunners control shall not be left unattended during immersion.) The gunners control shall meet the leakage requirements of 3.3.2.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10559221. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The gunners control is a functional unit of the Laser Range Finder AN/VVS-1. The gunners control contains indicator lamps (indicating operational conditions), logic selection switches and most of the control logic circuitry.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Applicable packaging data sheet number (see 5.1)

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(c) Selection of applicable levels of preservation, packaging, and packing
(d) Applicable stock number
(e) Requirement for First Article Submission

Custodian:

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Preparing activity

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